

Daman Dhaliwal

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Technical Skills

Programming & Data Systems: Python, SQL, R, Julia, TypeScript, Bash, Git, Docker, REST APIs, pytest

Analytics Engineering: Pipeline Design, Semantic Reporting Tables, Metric Definitions, Data Contracts, Partitioned Data Modeling, Parquet Panels, Power BI

ML & Causal Inference: Scikit-learn, Statsmodels, Panel Data Methods, Double ML, Causal Forests, IV, Sensitivity Analysis

AI Tooling & Business Translation: Gemini API, Browserbase Stagehand, Claude Code, Codex, LLM-Assisted Extraction, Model Evaluation, LLM-Readable Documentation, Scenario Modeling

Experience

Senior Financial Analyst, Chard | Vancouver, BC

June 2022 – July 2025

- Drove **\$1.0B** in institutional capital allocation decisions, building the financial and analytical models used to forecast asset performance and validate investment decisions for projects across Western Canada.
- Architected a market-intelligence pipeline using **Browserbase Stagehand**, **Gemini API**, **TypeScript**, and **Python** to extract municipal filings and combine them with structured market data.
- Converted spreadsheet-based financial models into **SQL-backed analytical systems** with **MS SQL** and real-time Sage accounting data, creating **semantic reporting tables** with standardized business logic for forecasting, variance analysis, and project health reviews.
- Managed the financial side of monthly and quarterly project health reviews by developing **Power BI** dashboards tracking **11 projects**, partnering with project managers, and briefing the **CEO, COO, and CFO** on financial performance, emerging risks, and forecast changes.
- Led the design and implementation of **data contracts** across project financial models, standardizing inputs, assumptions, metric definitions, forecast updates, and reporting outputs into a consistent source of truth for finance, development, and executive stakeholders.

Researcher, Journal of Political Economy, University of Chicago | Chicago, IL

July 2025 – Present

- Audited empirical research codebases by executing author-provided pipelines across **R, Julia, Python, Stata, and shell**, validating statistical methods, generated outputs, and causal-inference claims against manuscripts.
- Co-developed standardized replication workflows with **Bash, Git, Docker, and virtual machines**; documented method, dependency, and data-quality discrepancies in review-ready reports on reproducibility and empirical credibility.

Data Scientist, Sciences Po | Paris, France

January 2024 – December 2024

- Engineered a **real-time geospatial data pipeline** integrating **UK Government Transit APIs** and the **Open Source Routing Machine (OSRM)** to ingest, process, and store **50M+** GPS observations, enabling large-scale performance benchmarking across London's bus network.
- Built route-level efficiency measures by map-matching live bus locations to scheduled routes in **R**, classifying revenue service versus dead miles and quantifying spatial utilization at scale.

Selected Projects

PyFixest (718K Downloads) | Core Contributor, py-econometrics

github.com/py-econometrics/pyfixest

- Implemented new core features in a production-grade open-source analytics library: configurable demeaning iterations, **Driscoll-Kraay** and **Newey-West** standard errors, and sensitivity analysis functionality.
- Owned end-to-end feature development across **API design, numerical correctness, backward compatibility**, tests, documentation, and predictable behavior across edge cases.

Regime-Switching GP Asset Pricing | Python/JAX

github.com/damandhaliwal/Regime-Switching-GP-Asset-Pricing

- Built an **end-to-end forecasting pipeline** that converted raw **WRDS, Compustat, FRED, and VIX** data into point-in-time **Parquet** panels across **300 monthly cross-sections**, with reusable feature, prediction, and evaluation tables.
- Designed a regime-switching Gaussian process model with macro-driven state transitions to decompose market signals, compare forecast performance across regimes, and evaluate robustness in rolling out-of-sample tests.

Education

MA, Economics (Research Emphasis) | University of Toronto

May 2026

Coursework: Applied Causal Machine Learning (A+), Statistical Methods for Machine Learning and Data Mining (A-), Econometrics (A), Economics of Algorithms (A+), Computational Methods for Heterogeneous-Agent Models (A+)

B.Comm, Finance (with Honours) | University of British Columbia

May 2022

Award: Karen McKellin International Leader of Tomorrow Award (\$300,000 merit-based)